

Research Methods — Module 02: Agents — Study Questions

Recall and Comprehension

1. What is a Markov Blanket in empirical terms? How do you operationalize it for a biological system?
2. What are the key components of a generative model for agent behavior?
3. How do you estimate generative model parameters from behavioral data?
4. What is Dynamic Causal Modeling (DCM)? How does it relate to Active Inference?
5. What is computational phenotyping? How does it move beyond group-level analysis?

Analysis and Application

1. How do you dissociate model-based from model-free behavior using Active Inference?
2. How does the two-step task (Daw et al., 2011) reveal generative model structure?
3. What does interoceptive accuracy (heartbeat counting task) reveal about the precision of the self-model?
4. How do you measure the "depth" of an agent's generative model? What behavioral signatures distinguish shallow from deep models?
5. How does eye-tracking during social interaction reveal the observer's model of the agent they are watching?
6. What does sensory attenuation (force-matching task) tell us about efference copy and the self-model?
7. How do virtual reality manipulations of the rubber hand illusion inform us about the flexibility of the Markov Blanket?

Research Design

1. Design an experiment to test whether a particular brain region implements a component of the self-model. Specify: (a) the manipulation, (b) the imaging modality, (c) the Active Inference prediction.

2. How would you test whether an agent's behavior is better explained by Active Inference or expected utility theory? What is the key difference in their predictions?
3. You want to measure individual differences in precision on interoceptive signals. What physiological measures would you combine, and how would you control for confounds (e.g., BMI, medication)?
4. What ethical considerations arise when using computational phenotyping to classify individuals (e.g., for psychiatric diagnosis)?

Critical Evaluation

1. How does the concept of "agency" in Active Inference relate to philosophical debates about personal identity? Is the Markov Blanket sufficient to define an agent?
2. A study claims that a particular neural population "encodes" prediction error. What evidence would convincingly demonstrate this, and what would merely be consistent with it?
3. Critics argue that Active Inference is unfalsifiable because any behavior can be explained by choosing the right generative model. How do you respond?
4. Critically evaluate the use of heartbeat counting as a measure of interoceptive accuracy. What alternative measures exist, and what do they capture that heartbeat counting does not?
5. Design a longitudinal study to test whether precision on interoceptive prediction errors changes with mindfulness meditation training. Specify the controls and the duration.
6. How does cultural variation in body awareness (e.g., somatic vs. psychological idioms of distress) challenge a universal self-model account?